

A Reachability-Pruned Activity-Based Demand Model for MATSim

**Congestion Response and Demand Generation in
Higashi-Hiroshima**

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The Loop That's Usually Left Open

- A demand model feeds MATSim; assignment then produces the **congestion that demand model never saw** — a one-way street
- So the demand model optimizes in a **traffic-free world** — it never sees the congestion its own demand creates, and its behaviour answers to travel times nobody actually experienced
- Two things go untested: does the model's fit **survive contact with real congestion**, and can it even produce output MATSim can load **at full study-area scale**

An activity-based DDCM is a good test case: preferences and environment are cleanly separated, so MATSim's congestion can be handed back **without touching a single estimated parameter**.

Two Things We Wanted to Know

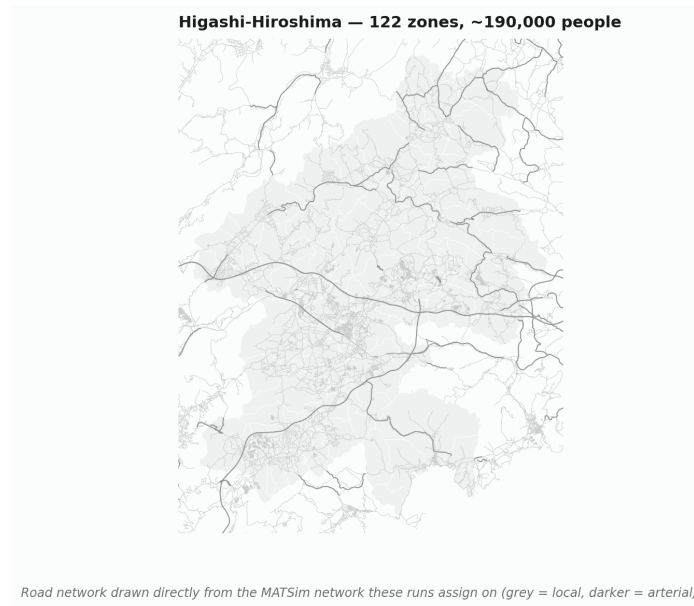
- **Does the coupling matter?** — supply MATSim's own congested travel times; do the observed diaries become **more probable** under them than under the table we estimated on?
- **Does it scale?** — can it write individually-optimized full-day activity-travel plans MATSim loads and assigns for a whole study area?

Parameters are **preferences** — structural, so they do not change with the environment. We estimate them once and hold them fixed. **What we feed back is the environment, never the parameters.**

The Point Behind the Two Questions

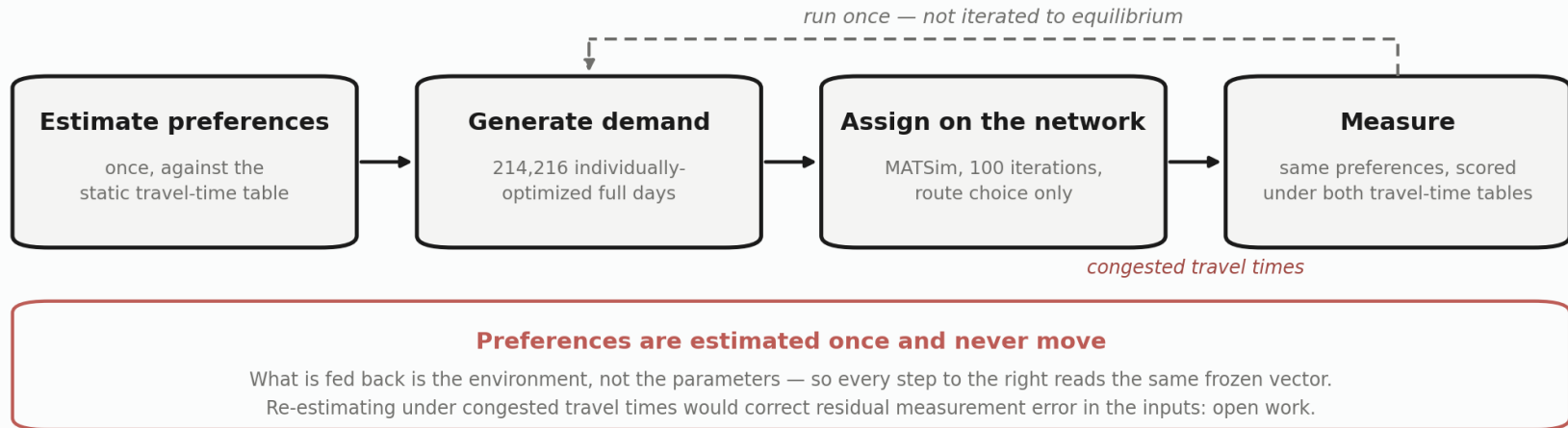
- An activity-based DDCM's real advantage over survey-matched demand: behavior is **solved**, not sampled — every agent optimizes their own day, so the population can respond to a scenario it has never seen
- That advantage only holds if preferences survive contact with the network conditions the simulator actually produces, not just the free-flow table they were estimated on — **that is Objective 1**
- It is only useful at policy scale if the model can write an entire city's worth of individually-optimized days — **that is Objective 2**
- Once both hold, one frozen θ^* can be pointed at any exogenous scenario — congestion, a lost car, a new transit line — and return a genuinely different, consistent population response

Higashi-Hiroshima



- ~190,000 people, 30 km east of Hiroshima City — Hiroshima University's move to Saijō made it a student city, with a sharply peaked station-to-campus commute
- Over 60% of trips go by car; bus is sparse and poorly tied to the JR Sanyō line
- Congestion concentrates on **National Route 2** at the university and work peaks — which is what the MATSim coupling has to reproduce

How the Two Models Connect



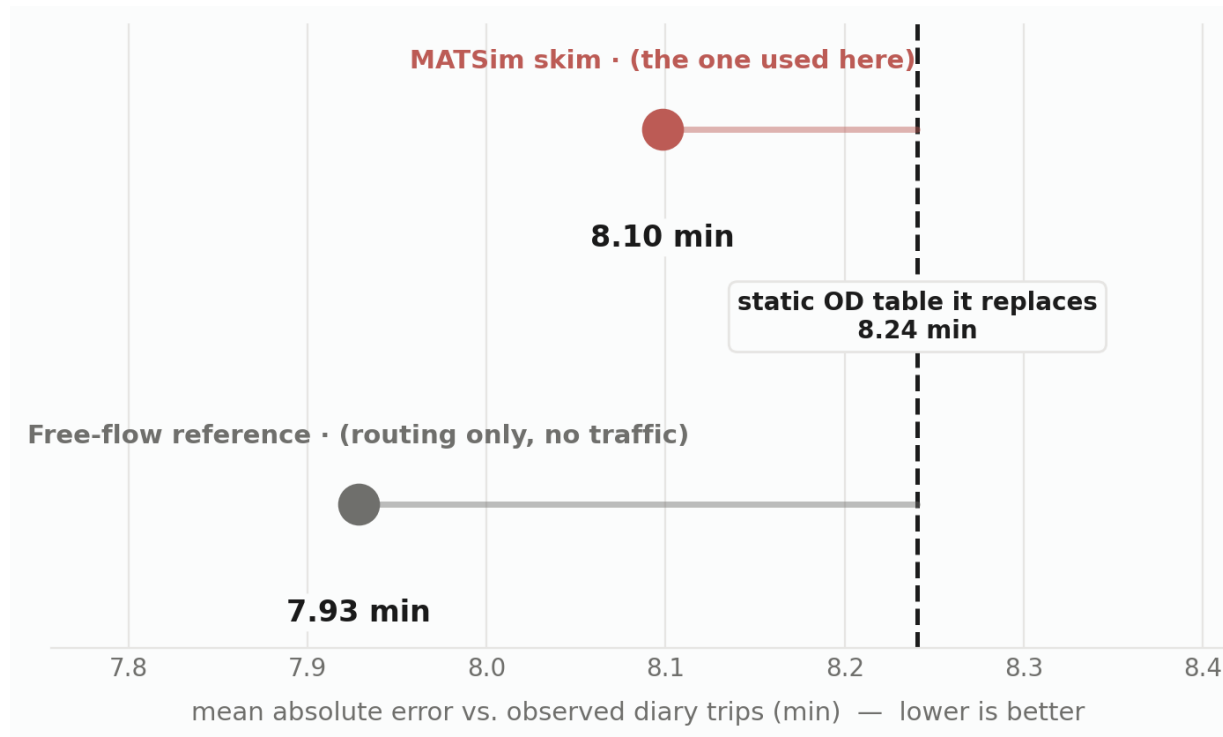
- **Preferences are estimated once and never move.** What travels back around the loop is the *environment* — congested travel times — not the parameters
- **Generation and assignment** are Objective 2; **the measurement** on the right is Objective 1
- The feedback ran **once**: everything downstream is a **first-order response**, not an equilibrium

One Parameter Vector, Two Travel-Cost Tables

- Evaluate the **same** $\hat{\theta}$ twice — against the static OD table, then against the skim
- Nothing is optimised; the difference in diary log-likelihood is the measurement
- The static arm reproduces our published likelihood to four decimals, so the harness is sound

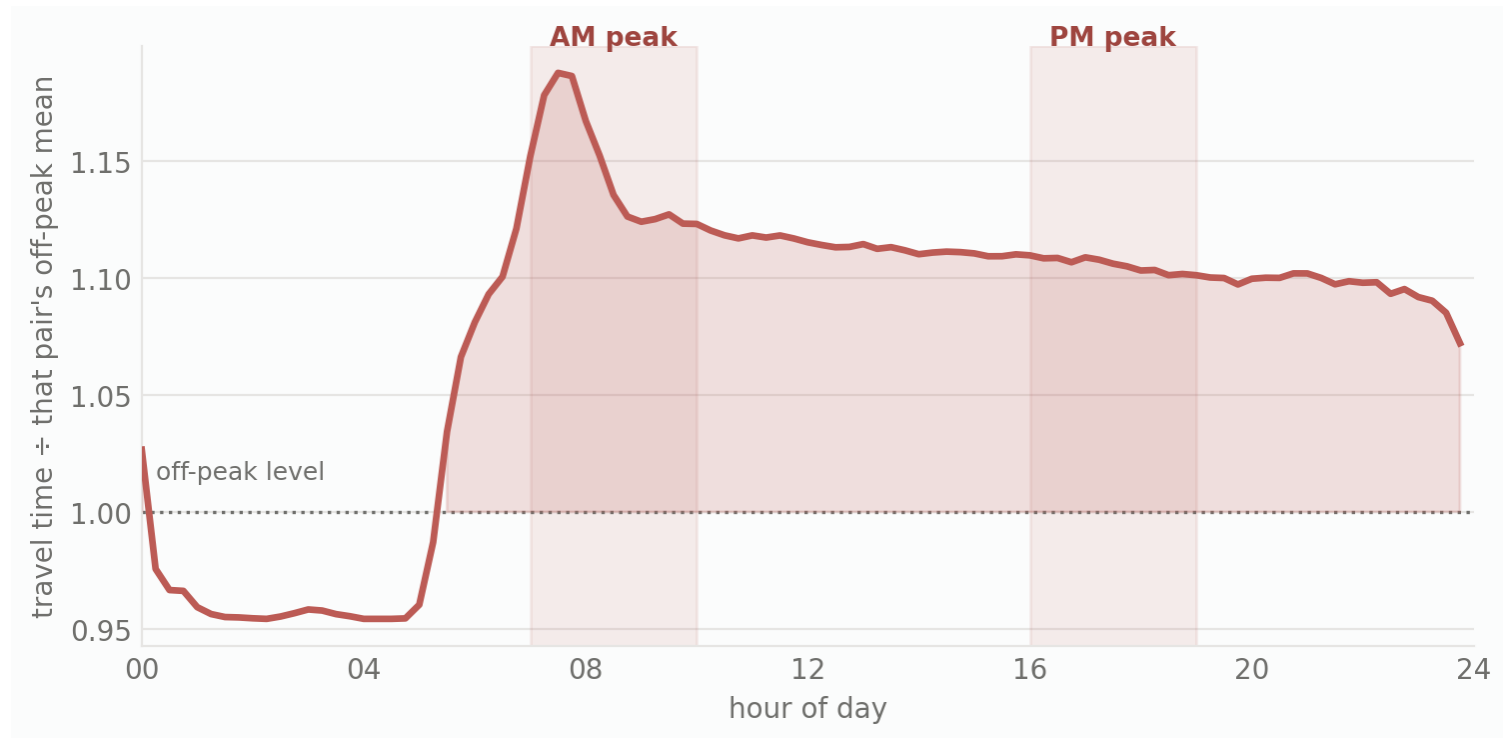
Travel times set arrival times, and arrival time is part of the state — so the skim arm rebuilds the graph, it does not merely reweight it.

First, Check the Skim Against Reality



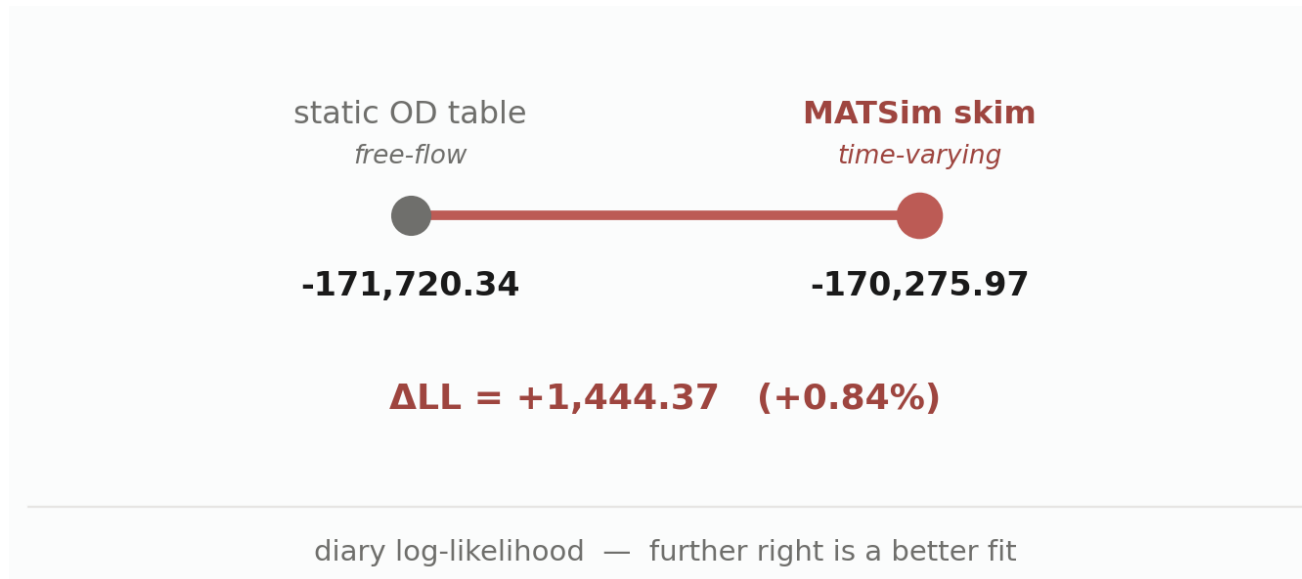
- The diary records real departures and arrivals — **observed travel time is observed**
- Our skim sits **1.7% better** than the table it replaces — **8.10 vs 8.24 min** against the diary's own observed travel times

Mild Congestion, but Real



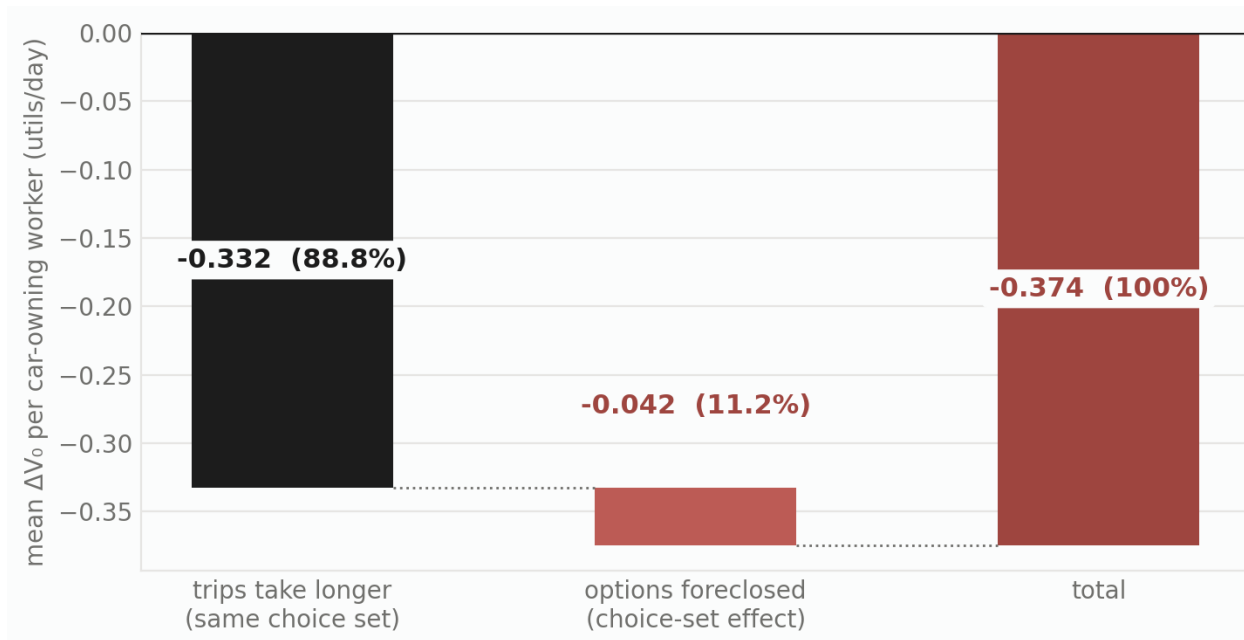
- AM peak $\approx 1.15\times$ the same pair's off-peak mean ($1.19\times$ at its busiest bin)
- A city of 200,000 — not a congested metropolis. This sets the scale of everything downstream.

Congestion Improves the Fit



- Same sample both arms · matched steps differ by **0.012%** — a clean comparison
- Modest, and proportionate to a network whose peak runs 1.15×

Mostly Delay, Not Foreclosure



- 88.8% trips simply taking longer · 11.2% options leaving the choice set
- Only a structural model can separate the two
- Worth knowing if you let MATSim replan: its mode and time changes **rescue** the agents losing options, which halves the foreclosure you would otherwise see

The Model Writes a Full Population

214,216

agents

45 min

to solve every day

4

shared graph topologies

2.65

trips/person (obs. 2.66)

The state-action graph **does not depend on preferences** — built once, reused. Only the utility array is rewritten per group.

MATSim Loads and Assigns It

214,216

DDCM day plans loaded

562,731

legs assigned on the real
network

98.95%

agents complete their day

0

warnings, full 100-
iteration run

- 157,475 agents actually travel that day (the rest stay home throughout) — of those, 1,653 (1.05%) get stuck and never complete their plan; everyone else does
- Route assignment only (next slide) — MATSim never touches mode or timing here, so this is purely: can it find every agent a path through the real road/transit network

MATSim Assigns, It Doesn't Choose

- θ has already chosen, for every one of the 214,216 agents: activities, destinations, **departure times and modes**. MATSim's default replanning re-decides the last two — `SubtourModeChoice` and `TimeAllocationMutator` — so **two mode-choice models compete over the same trips**
- Left alone, the congestion we hand back is partly *MATSim's* preferences, not DDCM's — the demand model's own output, returned to it as if it were an exogenous measurement
- Fix: the strategy config is cut to `ReRoute` (route choice) + `SelectExpBeta` (plan selector). MATSim finds each agent a path; it never re-decides **who** travels, **how**, or **when they set out**
- Checked, not asserted: mode share **byte-identical at iteration 0 and 100**, and every agent's **first departure unchanged** — 157,310 of 157,310. Later departures shift a little, because a faster route arrives earlier; that is the reroute propagating, not a decision

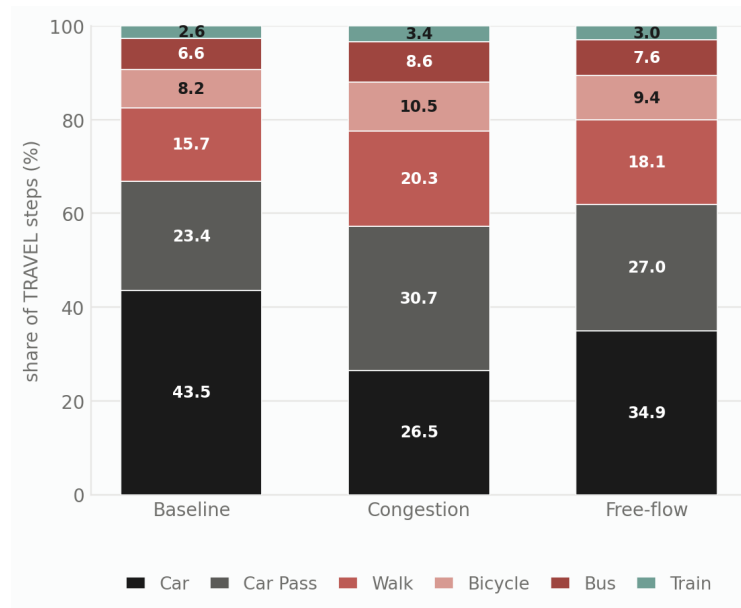
Congestion run: 100 iterations, flow/storage capacity 1.0 · one small caveat kept and documented rather than removed — an unseeded ± 10 min departure-time blur applied once at population load

What the Three Environments Are

- **Baseline** — the static OD table: an external routing engine, one number per zone pair, no time of day, never simulated
- **Congestion** — MATSim's own network, with the traffic our own population actually generates
- **Free-flow** — the same MATSim network, zero traffic: pure geometry

Congestion vs Free-flow is the clean comparison — same network, traffic on or off. Baseline sits on a different table: even at zero traffic MATSim's geometry runs **16.8% slower** than the routing engine, so reading Congestion against Baseline mixes congestion with that gap.

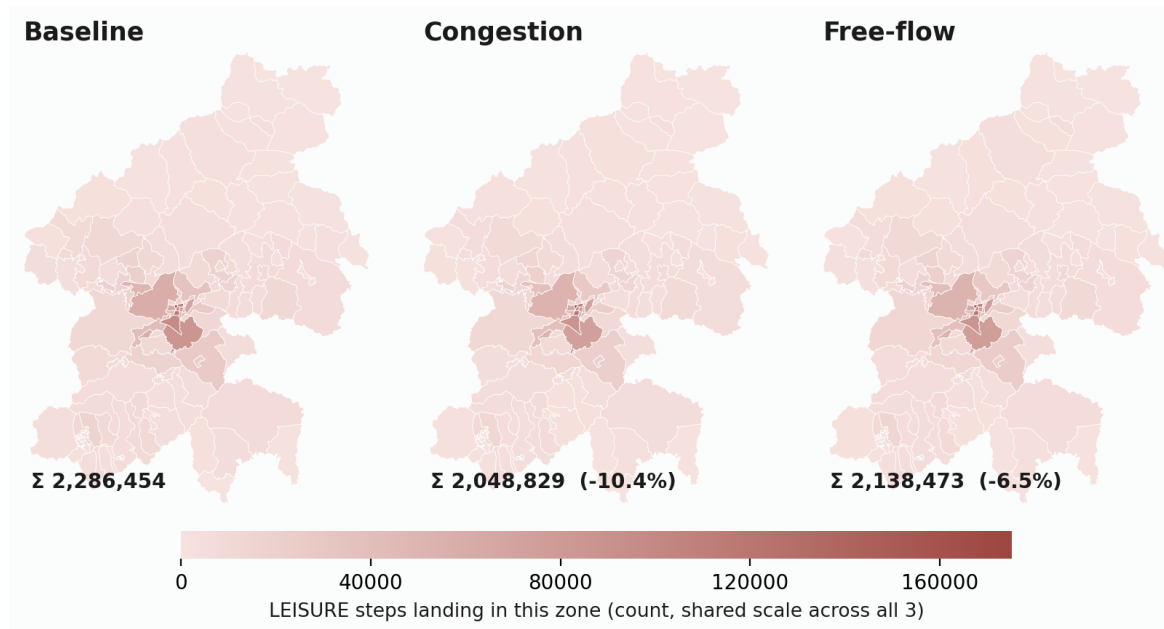
One θ , Three Environments



- Same frozen $\hat{\theta}$, three environments that all touch the real MATSim network — a baseline, real congestion, and its free-flow control — **no re-estimation, ever**
- Car share moves **34.9% → 26.5%** from free-flow to congestion — the clean comparison: same MATSim network, traffic on or off. Baseline's 43.5% sits on a different table (see notes)

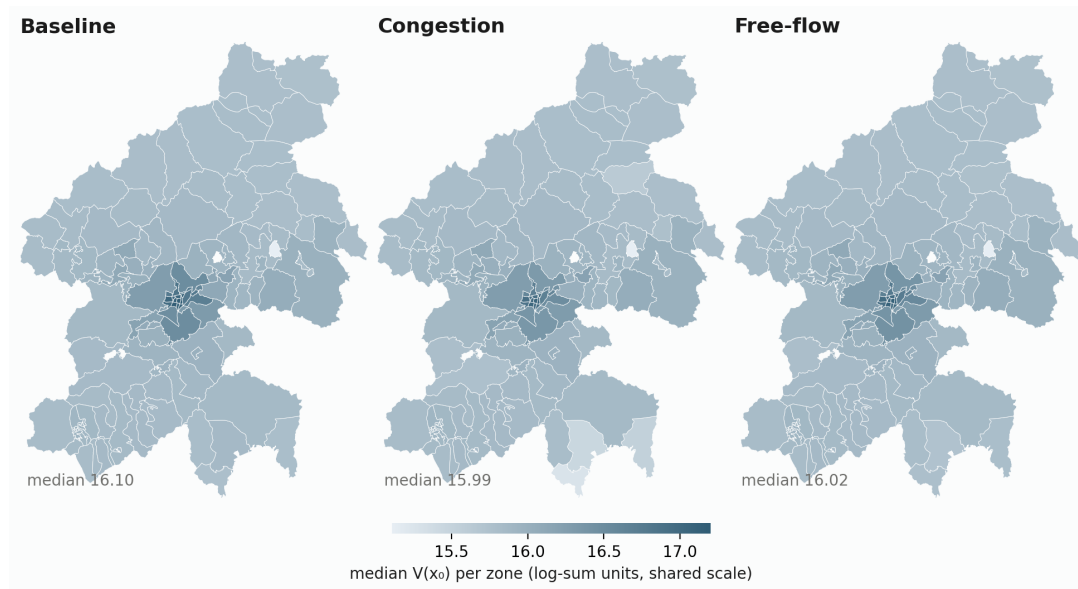
A **first-order** response, not an equilibrium — the loop was iterated once, and a fall in car demand this large would relieve some of the congestion that caused it

Same Places, Fewer Trips



- Same zone cluster stays darkest in every panel — destination choice barely moves whether congested or free-flowing
- What moves is volume (printed Σ on each panel): leisure activity drops **10.4%** under congestion relative to baseline, concentrating slightly into the most-attractive zone — trips are **foregone**, not relocated

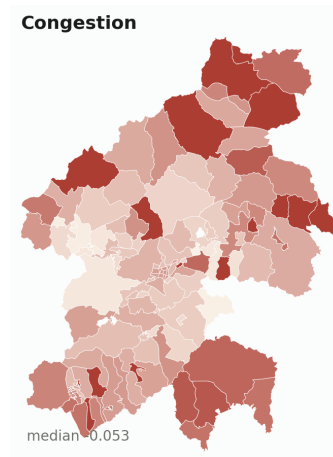
One Number MATSim Doesn't Give You



- Same three MATSim environments, now read as **accessibility** — $V(x_0)$, the log-sum over every feasible day, per person, per zone. Mode share and destination counts are both summaries of this one object.
- Looks modest for all three (16.10 → 15.99 → 16.02) — most residents aren't touched by any single environment, which dilutes the population-wide level.

Per-zone median, not mean

Where It Actually Lands



- **Congestion against free-flow** — the clean comparison, same MATSim network, traffic on or off. It touches **53.8%** of residents; **0 of 214,216** gain
- For those it touches, the **median person loses 0.203** log-sum units — about **5 minutes a day** of travel time. The worst-affected tenth lose over **9 minutes**
- The population median is only -0.053 , because **46% aren't affected at all** — which is exactly how averaging hides it

Per-zone median of the per-person change · against baseline instead it reads -0.140 , but that mixes in the $\sim 17\%$ network-methodology gap

What We Cannot Measure Well

- **Only car is simulated.** Bus and train come from the static table, unchanged, in every number here. MATSim's own PT times run **86 min median against 35 observed** — vehicles stuck at stops — so writing that level into a skim would bake a simulation defect in as a finding
- **And the table we fall back on is optimistic.** Against the diary's own bus trips it runs **17 min fast on 84.6%** of them, and makes bus quicker than *car* on **51.3%** of the pairs people actually used — impossible for a service that stops. The estimated bus constant has quietly absorbed that
- **Scaling that table by a ratio moves it without repairing it** — bus shifts -0.02 pp, the level's own error survives. And the ratio breaks across a network change, returning a 49% rail speed-up no bus corridor can cause
- **Train reaches 1.3% of zone pairs** — all the coverage the static table has
- **Our skim still loses to its own free-flow reference** — 8.10 vs **7.93** min against the diary, so the residual sits in how *traffic* is measured, not in routing

What the Model Cannot Represent

- **Service quality reaches the model only as time and cost.** There is no frequency, reliability or comfort term, and car's estimated constant sits **36 minutes** of travel time above bus. A new service can still be evaluated — the BRT in the appendix is, and people do switch to it — but only through the minutes it saves, which makes any such answer a **lower bound**
- **Preferences were fitted on free-flow inputs.** Re-estimating under congested times is owed — not because preferences change, they are structural, but because we used free-flow times as *input* where people faced congested ones. That is measurement error, and Objective 1 is what tells us how much it is worth correcting
- **One pass, not an equilibrium.** The feedback ran once, so every scenario number here is a **first-order** response

What We'd Like to Leave You With

- An activity-based DDCM **scales to a whole study area** — 214,216 agents in 45 minutes
- Coupled congestion **improves the fit**, in proportion to the congestion really there
- The point of both: one frozen θ is **reusable across environments** — no re-estimation between them
- **Validate the skim before you trust it.** The diary already knows how long trips took.

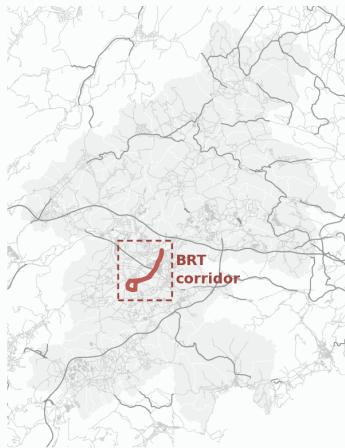
Extended abstract: MUM2026 #36 · m240492@hiroshima-u.ac.jp · azwanaza@gmail.com

Appendix

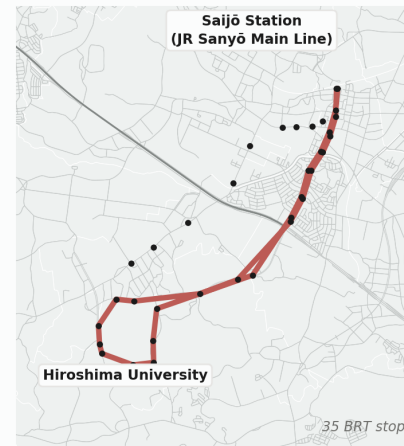
The BRT corridor — the intervention, the mechanism, and what it does

The BRT Proposal

Higashi-Hiroshima — 122 zones, ~190,000 people



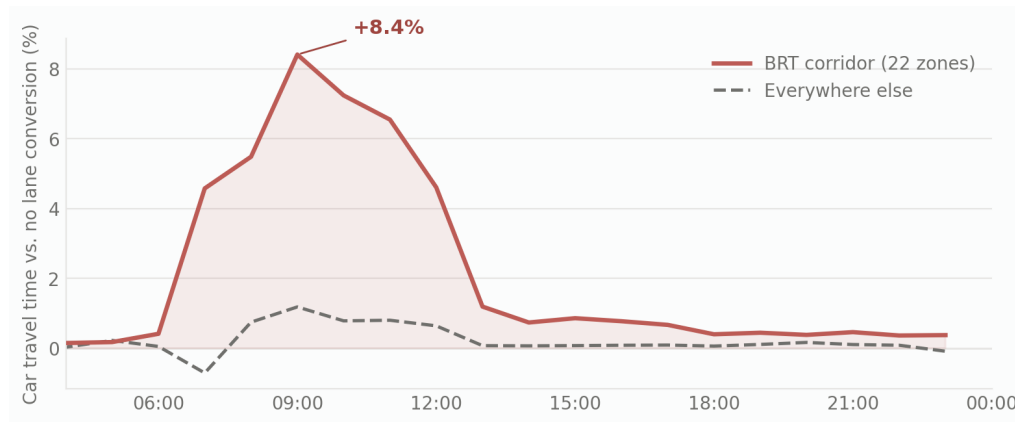
The proposed corridor — Saijō Station ↔ Hiroshima University



Road network drawn directly from the MATSim network used in these runs (grey = local, darker = arterial); corridor and stops from its transit schedule.

- Saijō Station ↔ Hiroshima University, 35 stops, ~5 km — BRT every 6 min at morning peak, 12 evening, 15 off-peak
- The proposal **converts one traffic lane each way** to a dedicated BRT lane, so road capacity for cars *falls* — two effects pulling against each other
- Corridor design from our group's companion WCTR study (Li et al., 2026), which evaluates it on a 2040 network; ours applied it to the current-year network

What the Lane Conversion Costs



- 301 corridor links converted in MATSim's own network — capacity **1900** → **950**, two lanes to one. Corridor car time rises **+8.4% at the 09:00 peak**, against **+1.2% elsewhere**
- DDCM's response: car share **26.5% → 26.1%**, mostly into car-passenger and walk. **50.6% of residents lose**, and corridor residents lose **6× the median** of everyone else

This is the lane conversion **in isolation** — a capacity cut with nothing running on the lane it freed. The service it exists for is on the next three slides, and it reverses the sign.

How the BRT Enters DDCM

- It has to enter through `bus_time`. DDCM decides who rides what, and it sees the world only through the travel-cost table. A BRT built in MATSim's network alone sits on the wrong side of the wall — correct lanes, correct routes, and invisible to the demand model
- **BRT is BUS, not a new mode.** A new mode needs an `asc_brt` we cannot estimate — there are no observations of a service that does not exist — and splitting one alternative into two near-identical ones inflates their combined share through IIA (red bus / blue bus)
- **We apply the difference, never the level.** Door-to-door under both timetables, then

$$\Delta = \min(0, [\text{access} + \text{wait} + \text{ride}]^{\text{BRT}} - [\text{access} + \text{wait} + \text{ride}]^{\text{No BRT}}) \cdot \text{bus_time} = \text{static} + \Delta$$

Both read from the transit timetable — an input file — never from MATSim's simulated PT times · No BRT = today's 2024 schedule

- **Why the difference and not the level?** Our bus table is **17 min optimistic on 84.6%** of observed bus trips, and the estimated bus constant already absorbs that. Replace the table without re-estimating and bus share collapses

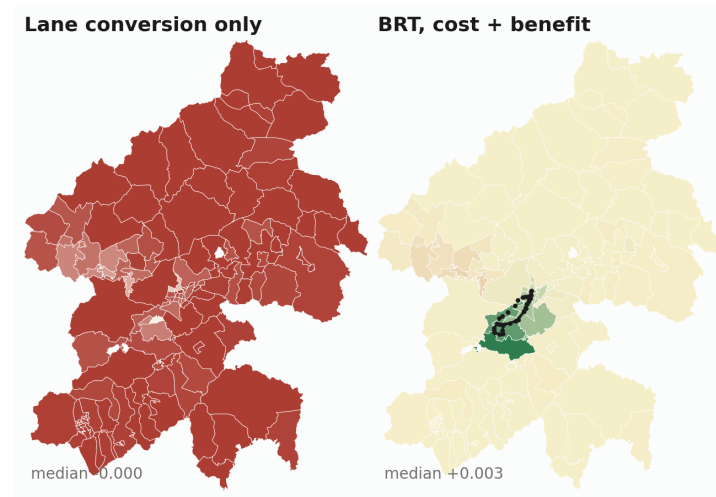
Measured Δ on the corridor: **-9.25 min** of waiting (24.5 → 6 min headway), **+1.2 min** of walking (43 BRT stops against 1,498 existing, so access gets worse). **67% of corridor cells move, 0.98% elsewhere.**

What the BRT Actually Does

- Bus **8.61% → 9.63%** (+1.02 pp), car **26.54% → 25.54%** (−1.00 pp), trips per person 2.400 → 2.415
- **People choose the service, not just away from the car.** Spread car's lost share evenly and bus would gain 0.12 pp; it gains **1.02 pp, nine times more** — while car-passenger, bicycle and train all lose ground
- Welfare **+0.0141 utils**. The lane conversion alone was **−0.0042**, so the service benefit outweighs its cost **4.4×**

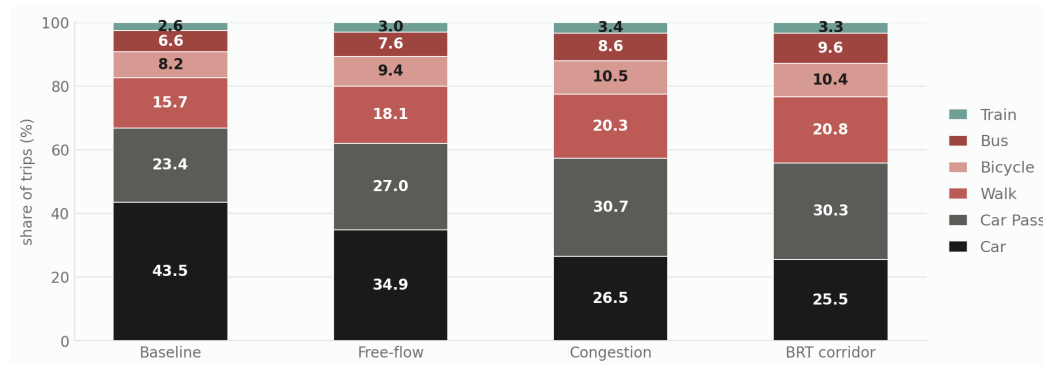
Without the service, the same lane conversion lifts every mode by almost exactly its existing share — people leaving a slower car, going wherever is next-best

The Corridor Gains, the Rest Pays a Little



- **Corridor residents: 96.6% gain, mean +0.0348** — the green band follows the actual route, drawn from MATSim's own transit schedule
- **Everyone else: -0.0014 on average, 45.6% lose.** They pay the lane conversion in extra car time without gaining access to the service
- Left panel: the same lane conversion **without** the service. Uniformly red — everyone pays a little, nobody gains

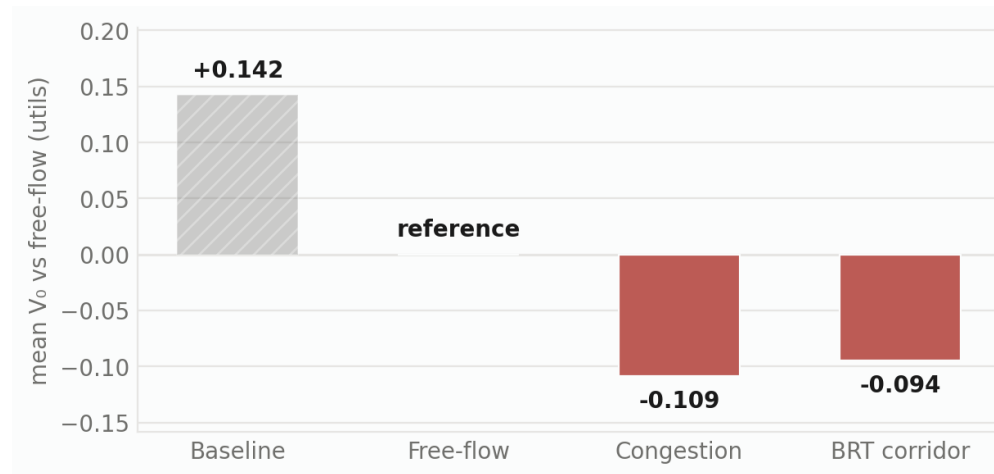
The BRT Next to the Other Environments



- Free-flow, Congestion and the BRT all sit on the **same MATSim network** and differ only in what is added: traffic, then a converted lane *and* the service that lane is for. Baseline sits on the static table, ~17% faster even at zero traffic — **not a clean difference** against the other three
- Congestion costs car **-8.35 pp**. The BRT corridor takes a further **-1.00 pp** off car — but unlike congestion, it puts it somewhere specific: **bus +1.02 pp**
- Congestion pushes car demand outward into every remaining mode in proportion. The BRT pulls it into one. That difference in *shape*, not size, is what separates a service from a penalty

Without the service, car's loss spreads evenly over every other mode. With it, the loss concentrates: bus takes **nine times** its even share, and three modes lose ground.

Read as a Log-Sum



- Plotted as a **change against free-flow**, not as levels: the levels are 15.40 / 15.25 / 15.15 / 15.14, and a level chart would need a truncated axis to show anything
- Congestion costs **-0.109 utils**; the BRT corridor gives back **+0.014**, about an eighth of it. Trips per person 2.496 → 2.400 → 2.415
- This is the number MATSim does not give you: a single welfare-consistent value per person, from the same $\hat{\theta}$, comparable across all four environments

Four Things This Number Leaves Out

- **The bus table's level is wrong by ~17 min**, and we did not fix it — we applied only what the policy changes, on the level θ^* was calibrated against. Repairing the level properly means re-estimating preferences, which is a different job
- **BRT in-vehicle time comes from the proposal's own timetable**, not from anything we measured. It is what its authors assume a dedicated-lane bus achieves
- **One BRT minute is valued exactly like one bus minute**. Reliability, comfort and legibility are not in our utility, so this is a **lower bound** on the BRT's appeal. Capturing them means adding attributes and SP data, not adding a mode
- **One pass, no re-assignment**. MATSim never moves the new bus riders back off the road, so the congestion they stop causing is not counted either

Two of these bound the result **downward**. A fuller treatment should make the BRT look better, not worse.